

Advanced Short Circuit Preventive Mechanism for Electric Vehicles

1. Theme

The project focuses on developing an advanced mechanism to prevent short circuits in electric vehicles (EVs) by incorporating real-time monitoring, fault detection, and preventive measures to ensure the safety and reliability of the electrical system.

2. Problem statement

Short circuits in electric vehicles can result in severe safety hazards, equipment damage, and operational downtime. Current preventive mechanisms often lack real-time detection and predictive capabilities, leading to delayed responses. The need is for a smart and efficient system that can predict, detect, and mitigate short circuits proactively to enhance EV safety, reduce maintenance costs, and improve performance reliability

3. College Code & College Name

College code : 9207

College Name: M. Kumarasamy College of Engineering.

4. Guide Name, Designation, Mobile No. & Email ID

Guide Name : Dr.K.Sheikdavood

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5. Student Team details

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6. Project Summary

The Advanced Short Circuit Preventive Mechanism for Electric Vehicles is a comprehensive solution aimed at enhancing the safety and reliability of EVs. The project focuses on designing a real-time monitoring system that can detect, predict, and prevent short circuits in the electrical system of an EV. The mechanism employs a combination of sensors, microcontrollers, and predictive algorithms to monitor critical parameters such as current, voltage, and temperature. Abnormal readings trigger alerts and activate preventive measures, such as isolating faulty circuits or adjusting power flow. The system leverages AI-driven analysis to predict potential faults, minimizing risks and enabling proactive maintenance. This innovative approach ensures enhanced safety, reduced repair costs, and improved vehicle lifespan, addressing a critical issue in the growing EV market.

7. Proposed solution with methodology

System Design and Component Selection:

- Utilize current, voltage, and temperature sensors to measure real-time parameters of the EV electrical system.
- Deploy a microcontroller (e.g., ESP32) as the central processing unit to handle data acquisition and communication.

Data Acquisition:

- Sensors continuously monitor key electrical parameters, including current spikes, voltage drops, and temperature surges, which are precursors to short circuits.
- Sensor data is transmitted to the microcontroller for further analysis.

Fault Detection:

- Define threshold limits for each parameter: current $> 120\%$ rated, voltage $\pm 10\%$ nominal, temperature $> 80^{\circ}\text{C}$.
- Employ an AI-based anomaly detection algorithm to identify deviations from normal operating conditions, predicting potential short circuits.

Preventive Actions(If anomalies are detected):

- Trigger alerts to notify users of potential risks.
- Isolate faulty circuits using relays or circuit breakers to prevent damage.
- Adjust power distribution to ensure continued operation without overloading other components.

8. Workplan / time schedule indicating the project mile stone

- Week 1-2: Perform requirement analysis, finalize design specifications, and select components (sensors, microcontroller, relays).
- Week 3-4: Design the circuit, procure materials, and assemble the hardware system.
- Week 5-6: Implement AI-based fault detection algorithms and integrate with the hardware.
- Week 6-7: Test and validate the system through simulations, fine-tuning thresholds, and optimizing performance.
- Week 7-8: Develop the real-time monitoring dashboard, finalize integration, document findings, and prepare the project for demonstration.

9. Plan of action of implementation

- Analyze requirements, select components such as current sensors (ACS712), voltage sensors (ZMPT101B), temperature sensors (LM35), relays, and ESP32 microcontroller, and finalize the design.
- Design the system, procure components like PCB boards, connectors, wires, and power supply units, and assemble the hardware, including the sensors and microcontroller.
- Implement AI-based fault detection algorithms, integrate with hardware components (ESP32), and perform testing.
- Optimize the system, develop a real-time monitoring dashboard, document findings, and prepare for the project demonstration

10. List of facilities available in the college to develop the prototype of the project

- Embedded System Lab
- Circuit assembly and microcontroller programming in an electronics lab.
- Testing facilities for prototype assessment.

11. Total Cost

The total estimated cost for the project is **₹10,000**.

12. Details of Financial assistance required

The financial assistance required is ₹10,000, which will be allocated for procuring essential components such as sensors, microcontrollers, relays, PCB fabrication, and testing equipment to ensure successful project implementation.

13. Expected outcomes / results

- **Functional Prototype:** A working system that detects and prevents short circuits in electric vehicles using real-time monitoring.
- **Enhanced Safety:** Proactive fault detection reduces risks of electrical failures and hazards.
- **User-Friendly:** Automated detection and preventive mechanisms require minimal user intervention.
- **Cost-Effective:** Designed with affordable components to ensure feasibility and scalability.
- **Environmental Benefits:** Prevents electrical fires and energy wastage, contributing to sustainable EV operations.
- **Adaptability:** Can be integrated into various EV models and electrical systems.
- **Awareness and Adoption:** Promotes the importance of advanced safety systems in electric vehicles.

UNDERTAKING

1. ALL the students are studying in final year engineering. All the students are registered only once for this scheme.
2. The college will provide the basic infrastructure and other required facilities to the students for timely completion of their projects.
3. The college assumes to undertake the financial and other management responsibilities of the project. We are aware that the amount is to be utilized only for the purpose sanctioned i.e. to meet the expenses for developing the prototype and not for purchase of computer consumables, stationaries, honorarium, overhead etc. Unutilised balance amount will be returned back to the University after the time of completion of the project.

Signature
(SHANJAY RK)

Signature
(SIVARAMAKRISHNAN M)

Signature
(SUBASH P)

Signature
(YASAR S)

Signature of the Mentor
(Dr.K.Sheikdavood)

Signature and seal of the principal
(Dr. B.S.Murugan,M.Tech,Ph.D)